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标题	FLP‐Catalyzed Transfer Hydrogenation of Silyl Enol Ethers
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摘要（原文节选）

AbstractHerein we report the first catalytic transfer hydrogenation of silyl enol ethers. This metal free approach employs tris(pentafluorophenyl)borane and 2,2,6,6‐tetramethylpiperidine (TMP) as a commercially available FLP catalyst system and naturally occurring γ‐terpinene as a dihydrogen surrogate. A variety of silyl enol ethers undergo efficient hydrogenation, with the reduced products isolated in excellent yields (29 examples, 82 % average yield). ⋯

数据表 · Table 1（16 行，含数值行 16 行）

Optimization of the FLP‐catalyzed transfer hydrogenation.[a]

Entry	Variation from “standard” conditions	Yield[b] [%]
1	none	96
2	no B(C6F5)3	<2
3	no TMP	<2
4	B(2,4,6‐F3C6H2)3 instead of B(C6F5)3	7
5	B(2,6‐F2C6H3)3 instead of B(C6F5)3	<2
6	DABCO instead of TMP	41
7	PMP instead of TMP	36
8	t‐Bu3P instead of TMP	43
9	2 b instead of 2 a	23
10	2 c instead of 2 a	<2
11	2 d instead of 2 a	<2
12	benzene instead of toluene	46
13	[1]=0.32 m	87
14	60 °C	<2
15	2 h	47
16	5 mol % catalyst	23

脚注（逐字）：[a] Reactions performed using 0.5 mmol of silyl enol ether 1 where [1]=0.16 m in toluene. [b] Determined by 1H NMR analysis of the crude reaction mixture with 1,3,5‐trimethylbenzene as the internal standard.

正文对应句（逐字）：To commence our studies, we selected silyl enol ether 1 as a model substrate (Table 1). After extensive optimization,13 it was found that a FLP system composed of B(C6F5)3 (10 mol %) and 2,2,6,6‐tetramethylpiperidine (TMP) (10 mol %) using γ‐terpinene 2 a (1.3 equiv) as a dihydrogen surrogate in tol

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